



News Recommendation Utilizing Diverse News Characteristics

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- User Encoder & Click Predictor
- Datasets
- Key Challenges in News Recommendation

❖ IP2

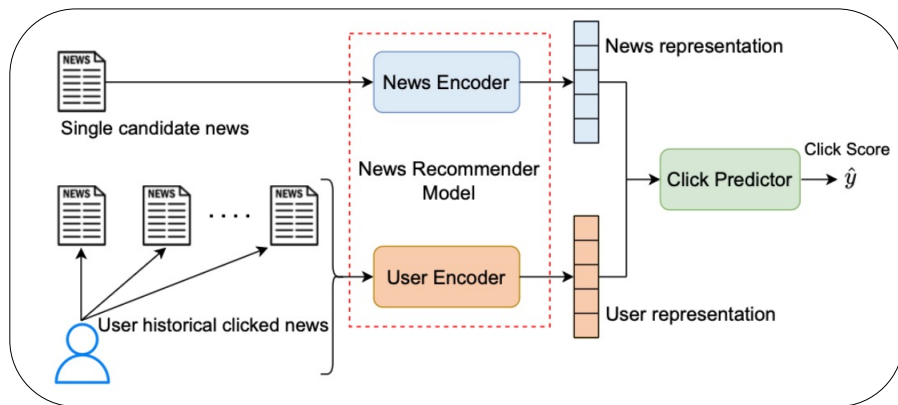
❖ GPM-LPM

News Recommendation

❖ Goal of News Recommendation

- Predict which news articles that the user is most likely to read

❖ Typical Approach



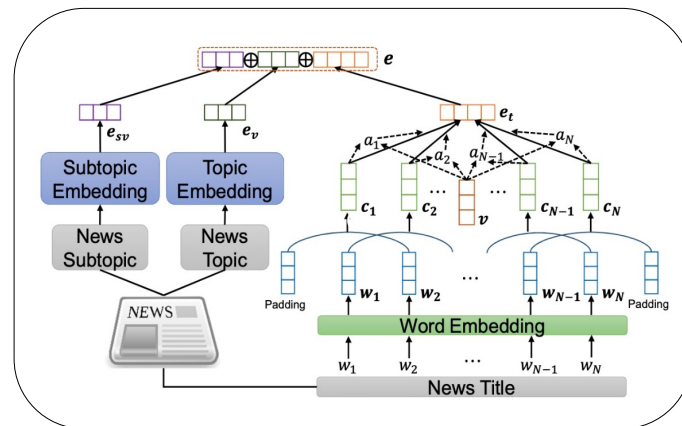
[1] News Recommendation Procedure

- Model the user's interests **based on the news that user has clicked on**
- Recommend the **candidate news** that is most related to the modeled user interest

News Encoder

❖ Represent News with Rich Information

- ❑ Text-based Information
 - Title, Abstract
- ❑ Metadata (Time-based features)
 - Published time, Interaction time
- ❑ Topic information
 - Category, Subcategory



[1] News Encoder using Title, Category, Subcategory

❖ Importance of News Embeddings

- ❑ Serve as core inputs for building the user embedding
- ❑ Also used to represent candidate news articles

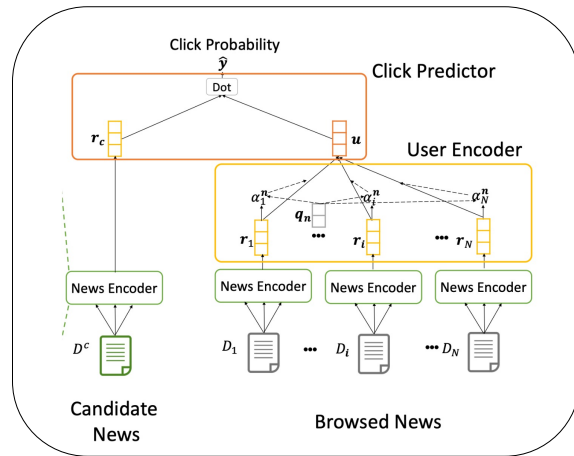
User Encoder & Click Predictor

❖ Build User Embeddings from News Embeddings

- ❑ Assign attention weights to clicked news
- ❑ Aggregate weighted news embeddings into a user embedding

❖ Estimate Click Probability

- ❑ Encode candidate news with the same news encoder
- ❑ Compute click probability
by comparing user and candidate news embeddings



[1] User Encoder and Click Predictor

Dataset (Adressa)



❖ User-Clicked News Pair Table

- ❑ Each event means that one user clicked one news article
- ❑ Contains **rich time-aware information**
 - event-occur time, activeTime, news publishTime
- ❑ News content-based information
 - Title, Body
 - Category, Keywords, Entities

Attribute	Description	Example
eventID	Id of reading event	1082287123 (integer)
Time	The time of the event	1487572383 (Unix time)
sessionStart	Indicates if the event is the first event in the session	False
sessionStop	Indicates if the event is the last event in the session	False
activeTime	The active time spent on a page	23 (seconds)
canonicalURL	URL of the visited page	"http://adressa.no"
referrerURL	The URL of the referrer page	"www.facebook.com"
documentId	Internal ID of page	"9757814edc2d346dfcf6f54e349f404c4e9775cf"
Title	Title of the article	"Test av 19 grovbrød"
Category	News category	"sport"
wordCount	Number of words on page	545 (words)
publishTime	Date of publication	"2017-02-20T09:45:47.000Z"
userID	The cross-site user identifier	"cx:i8i85z793m9j4yy0:cv8ghy3v45j8"
City	City inferred from user's IP address	"Verdal"
Region	Region inferred from user's IP address	"Nord-Trøndelag"
Country	Country inferred from user's IP address	"no"
os	Operating system	
deviceType	The type of device	

Title: "RBK finansieres av internasjonale spillerskaper" (RBK is funded by international betting companies)

Author: "Morten Klein"

Category: "meninger" (opinions)

Keywords: RBK, Byåsen håndball, spillemidler, sponsor

Entities: football club RBK (Rosenborg), RBK director Tove Moe Dyrhaug as well as a number of other clubs and some betting companies.

wordCount: 545 words

Body: «Det er med undring at jeg leser uttalelsene til RBKs daglige leder Tove Moe Dyrhaug i Adresseavisen 4. februar. Moe Dyrhaug uttaler seg i forbindelse med støtten på 1 million kroner undertegnede og Norgesspill.com har valgt å gi til Byåsen Håndball Elite. ...»

Dataset (MIND)

❖ Impression-aware interaction Table

□ Impressions

- ▶ **A sequence of news articles shown to a user at a specific time**
 - Provides time information for each impression
 - Clicked items are labeled as 1, and non-clicked items are labeled as 0

□ History

- ▶ IDs of news articles the user has clicked up before the current moment

□ News content-based information

- ▶ Title, Abstract
- ▶ Category, Subcategory, Entities

Column	Content
News ID	N37378
Category	sports
SubCategory	golf
Title	PGA Tour winners
Abstract	A gallery of recent winners on the PGA Tour.
URL	https://www.msn.com/en-us/sports/golf/pga-tour-winners/ss-AAjnQjj?ocid=chopendata
Title Entities	[{"Label": "PGA Tour", "Type": "O", "WikidataId": "Q910409", "Confidence": 1.0, "OccurrenceOffsets": [0], "SurfaceForms": ["PGA Tour"]}]
Abstract Entities	[{"Label": "PGA Tour", "Type": "O", "WikidataId": "Q910409", "Confidence": 1.0, "OccurrenceOffsets": [35], "SurfaceForms": ["PGA Tour"]}]

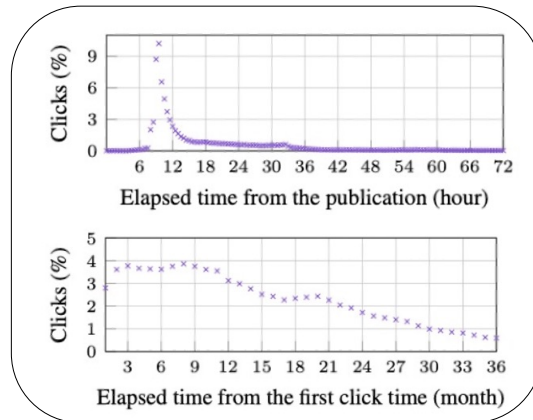
Column	Content
Impression ID	91
User ID	U397059
Time	11/15/2019 10:22:32 AM
History	N106403 N71977 N97080 N102132 N97212 N121652
Impressions	N129416-0 N26703-1 N120089-1 N53018-0 N89764-0 N91737-0 N29160-0

Key Challenges in News Recommendation

❖ Characteristics of News Items

❑ Highly time-sensitive

- ▶ Consumption spikes shortly after publication and fades rapidly
- ▶ Show distinct consumption patterns compared to items like movies



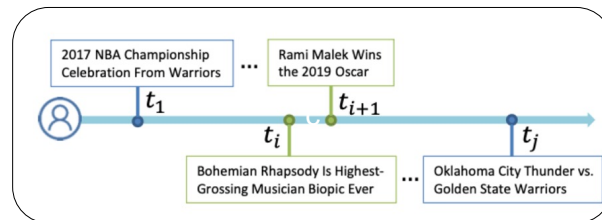
[1] Overall click rate time distribution
(Up: News, Down: Movie)

❖ User Interests are Diverse

❑ A user may have preferences across multiple categories

❑ User interests include both **long-term** and **short-term** preferences

- ▶ e.g., sports news during the Olympics
- ▶ e.g., political news during an election season



[2] Long-term preference, short-term interests

[1] Bae, H.-K., Ahn, J., Lee, D., & Kim, S.-W. (2023). LANCER: A Lifetime-Aware News Recommender System. Proceedings of the AAAI Conference on Artificial Intelligence, 37(4), 4141-4148.

[2] Mingxiao An, Fangzhao Wu, Chuhan Wu, Kun Zhang, Zheng Liu, and Xing Xie. 2019. Neural News Recommendation with Long- and Short-term User Representations. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pages 336-345, Florence, Italy. Association for Computational Linguistics.

Upcoming Topics

❖ Interest guidance led by entities in titles

- ❑ Specifically model the role of entities in the user's news selection process
- ❑ Intra-news and inter-news level interest guidance

❖ Using a time-aware user-news graph

- ❑ Divide the timeline into several time stages and build subgraphs
 - Capture overall interactions between users and news in each time stage
- ❑ Model how user interests evolve over time



IP2: Entity-Guided Interest Probing for Personalized News Recommendation

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Published in Recsys' 2025

Entity in News Recommendation Data

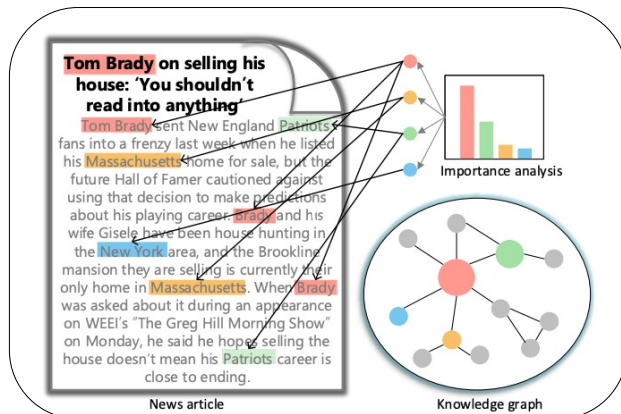
❖ MIND Dataset

Column	Content
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SubCategory	golf
Title	PGA Tour winners
Abstract	A gallery of recent winners on the PGA Tour.
URL	https://www.msn.com/en-us/sports/golf/pga-tour-winners/ss-AAjnQjj?ocid=chopendata
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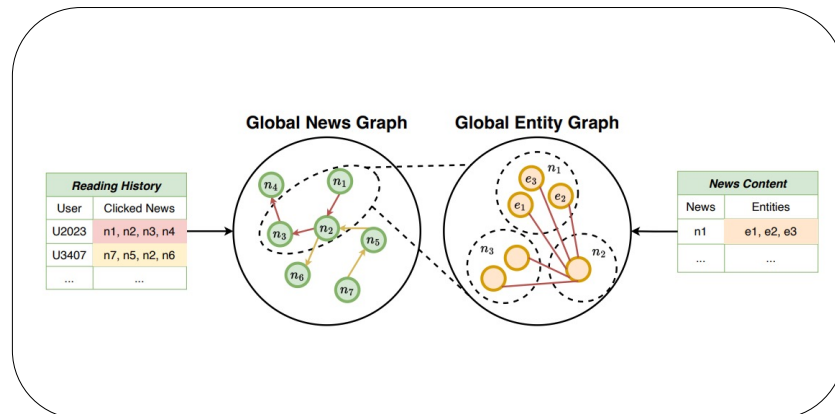
- ❑ Extract **entity information** from news titles using NER and entity linking

Conventional Entity Utilization Methods

❖ Limited Exploitation of Entities in News Recommendation



[1] KRED (Recsys' 20)

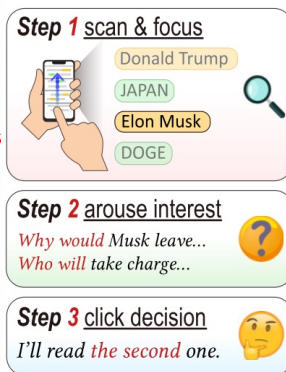


[2] GLORY (Recsys' 23)

- ❑ Integrate entity embeddings learned from knowledge graphs
- ❑ Use relationships between entities
- ➔ **Treat entities as mere semantic features, overlooking their unique roles**

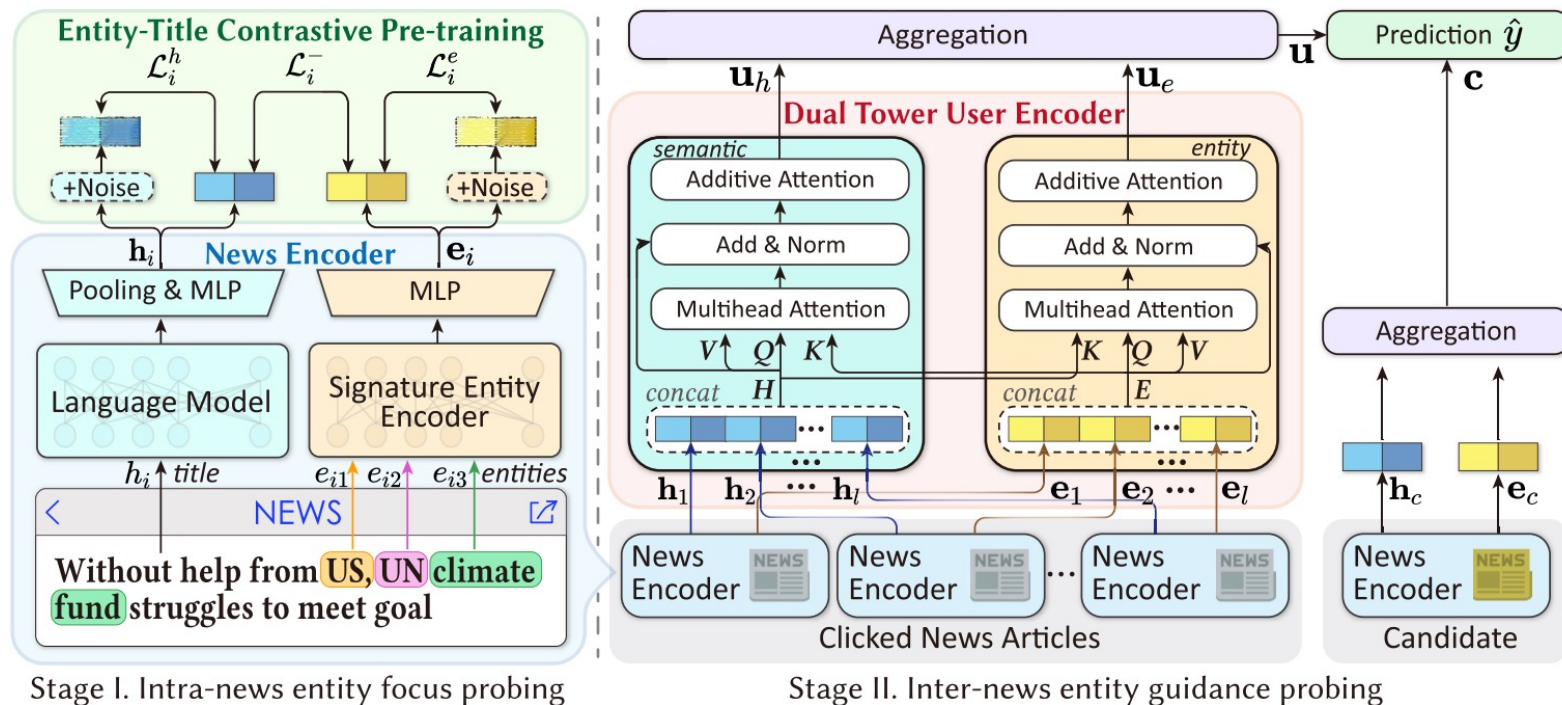
Unique Role of Entities in News Recommendation

❖ Scanning, Keywords Spotting and News Selecting



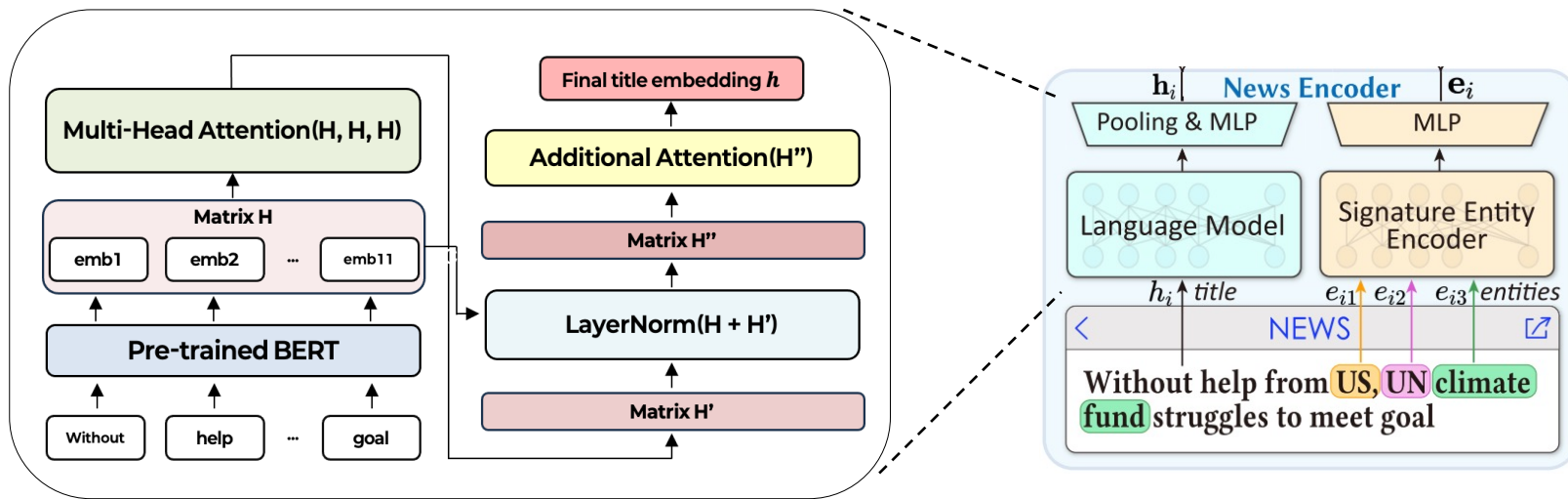
- ❑ At the **intra-news level**, a **leading entity** that would be the most attractive one during scanning (Musk)
 - Other **assistant entities** would help to emphasize the interest in this leading entity (DOGE)
- ❑ At the **inter-news level**, there's an **inter-news entity guided interest stream** (Trump → Musk)
 - That interest stream will influence **title reading**

IP2 (entity-guided Interest Probing for Personalized news recommendation)



News Encoder

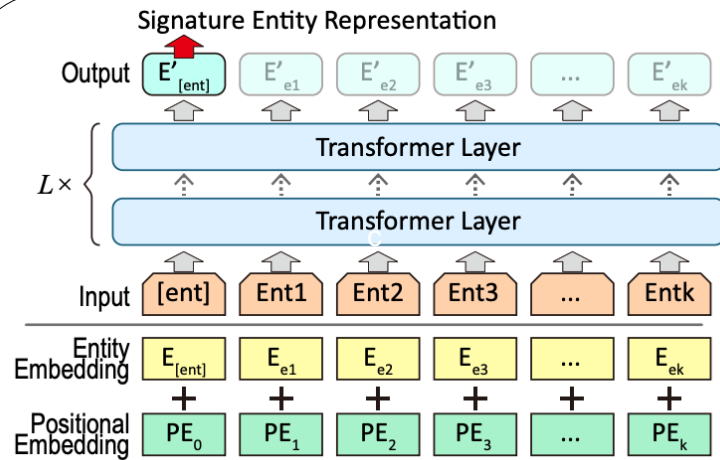
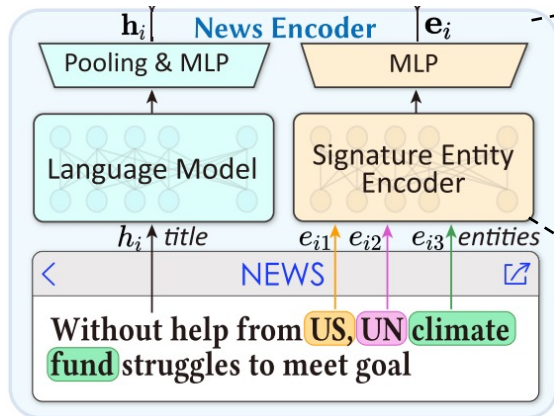
❖ Text-based Title Encoder



- ❑ Capture token-level semantics and intra-title token relationships
- ❑ Employ attention pooling to generate the final title embedding
 - Emphasize the unique contribution of each word

News Encoder (Cont.)

❖ Signature Entity Encoder (SEE)

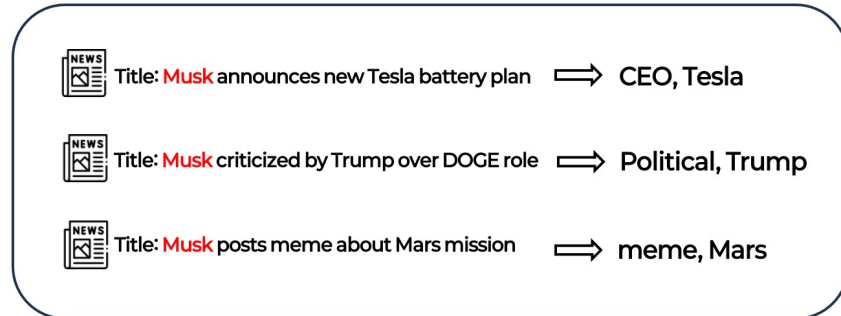
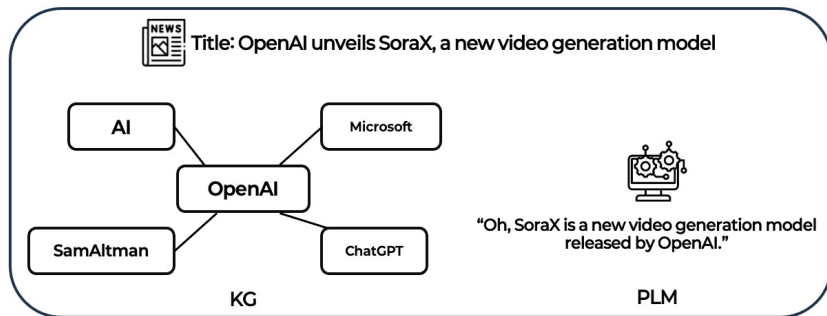


- ❑ Use **multiple Transformer Layer** to capture each entity's contextual importance in all co-occurring
- ❑ Employ **positional embeddings** to preserve the entity presence order
- ❑ Utilize the **[ent] position output** as the final signature entity representation
 - $[ent]$ serves as a [CLS]-like token in BERT, does not representing a real-word entity token

Learning Entity Embeddings without KG

❖ Initialize Entity Memory through Entity-Title Contrastive Pre-training

- ❑ Avoid using external knowledge graphs as knowledge sources
 - KGs suffer from sparsity, making newly emerging entities hard to link
 - Learning KG embeddings is **time-consuming** and **vulnerable to information staleness**
- ❑ Initialize entity memory through entity-title contrastive pre-training
 - Use **PLM-based title embeddings** as semantic signals and train entity embeddings to match the title context
 - Better handle newly emerging entities and changing title contexts



Signature Entity-Title Contrastive Pre-training

❖ Contrastive Learning within a Mini-batch

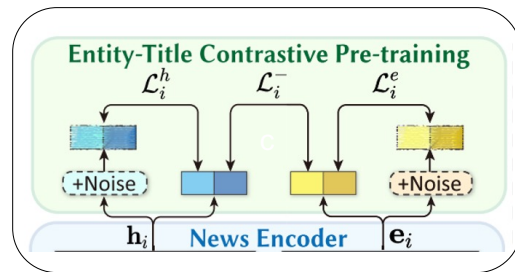
- For each news in mini-batch, obtain two embeddings from the news encoder

- h_i : title embedding, e_i : signature entity embedding

- Align each entity embedding with its corresponding news title

- Pull e_i closer to h_i , while pushing it away from other title embeddings

- $\mathcal{L}_i^- = -\log \frac{\exp(\text{sim}(\mathbf{e}_i, \mathbf{h}_i)/\tau)}{\sum_{j=1}^v \exp(\text{sim}(\mathbf{e}_i, \mathbf{h}_j)/\tau)}$



❖ Prevent Title and Entity Embeddings from Becoming Homogenized

- Encourage each embedding to stay close to its own “mirror” embedding

- Apply dropout layers as noisy gates to slightly modify the embedding

- $\mathcal{L}_i^h = -\log \frac{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau)}{\sum_{j=1}^v \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)/\tau)}$, $\mathcal{L}_i^e = -\log \frac{\exp(\text{sim}(\mathbf{e}_i, \mathbf{e}_i^+)/\tau)}{\sum_{j=1}^v \exp(\text{sim}(\mathbf{e}_i, \mathbf{e}_j^+)/\tau)}$

Dual Tower User Encoder with Cross Attention

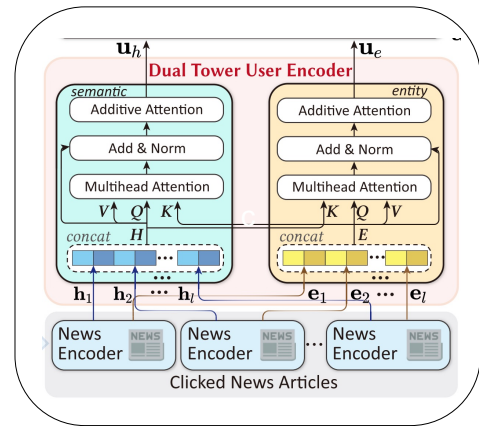
❖ Attention-based User Encoder with Two Identical, Parallel Towers

- ❑ Model semantic reading interests from news titles and entity-based reading interests
- ❑ Capture both long-term and short-term interests within a user's click sequence

❖ Cross attention capturing an interaction between two streams

- ❑ The initial entity interest will stimulate the user to read the title intensively
- ❑ In turn, spark curiosity about other entities

$$\begin{aligned} U_h &= \text{MHAttention}(\overset{\text{Key: Entity embeddings}}{\mathbf{H}, \mathbf{E}, \mathbf{H}}) & U_e &= \text{MHAttention}(\overset{\text{Key: Title embeddings}}{\mathbf{E}, \mathbf{H}, \mathbf{E}}) \\ \widetilde{U}_h &= \text{layernorm}(\mathbf{H} + U_h), & \widetilde{U}_e &= \text{layernorm}(\mathbf{E} + U_e), \\ \mathbf{u}_h &= \text{Add.Attention}(\widetilde{U}_h), & \mathbf{u}_e &= \text{Add.Attention}(\widetilde{U}_e), \end{aligned}$$



Aggregation of Two User Embeddings

❖ Different Users may Balance these Two Preference Streams Differently

- ❑ Most users may **directly skip title reading**, If they are **not interested** in the associated entities
- ❑ But there are users who may still take a glance

❖ Model user-specific balance between title semantic and entity preference streams

- ❑ A weighted sum of u_h and u_e
- ❑ Obtain the final user embedding u using a learnable weighting parameter

$$\begin{aligned} \mathbf{u} &= \eta_h \mathbf{u}_h + (1 - \eta_h) \mathbf{u}_e, \\ \eta_h &= \sigma([\mathbf{u}_h; \mathbf{u}_e] \mathbf{W}_a + b_a), \end{aligned}$$

Model Training

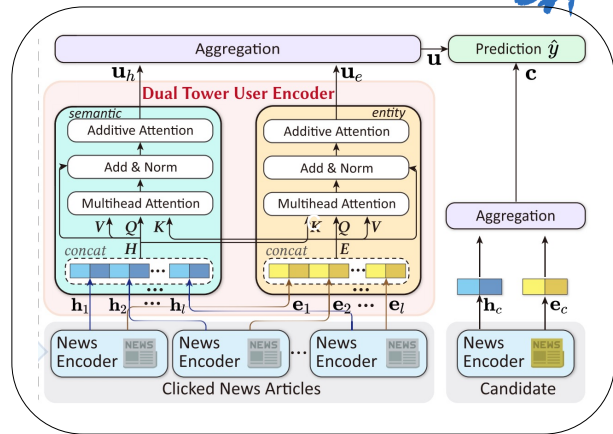
❖ Pre-training the News Encoder

- Train the news encoder with entity-title contrastive learning
 - $\mathcal{L}_i = \alpha \mathcal{L}_i^- + \beta \mathcal{L}_i^h + \delta \mathcal{L}_i^e$.
 - Export the news encoder into a checkpoint at the end of the last epoch

❖ Downstream Recommendation Training

- Obtain title and entity embeddings for each candidate news using the pre-trained news encoder
 - Aggregate title and entity embeddings using the same learnable weighting strategy as the user encoder
 - $\mathbf{c} = \eta_c \mathbf{h}_c + (1 - \eta_c) \mathbf{e}_c$,
 $\eta_c = \sigma([\mathbf{h}_c; \mathbf{e}_c] \mathbf{W}_a + b_a)$.

$$\square \quad \mathcal{L}_{NCE} = - \sum_{i=1}^{|S|} \log \frac{\exp(\hat{y}_i^+)}{\exp(\hat{y}_i^+) + \sum_1^r \exp(\hat{y}_i^{j-})}. \quad \hat{y}_i = \text{softmax}(\mathbf{u} \cdot \mathbf{c}_i),$$



Experiment

❖ Experimental Setup

- ❑ MIND (6 weeks)
 - Negative sampling on given “impression”
- ❑ Adressa-1 week
 - ~6 days: train data / last day: test data
 - Because no “impression” in Adressa, randomly sampling negative

❑ Data Preprocessing

- Drop all history logs that are shorter than 5,
 - truncate at 50 if longer than 50
- Limit the title length to 20 words

❖ Metrics

- ❑ AUC, MRR, NDCG@5, NDCG@10

Table 1: Dataset Statistics.

	MIND-small	MIND-large	Adressa
#users	94,057	1,000,000	640,503
#news	65,238	161,013	20,428
#entities	31,451	42,628	98,596
#clicks	347,727	24,155,470	3,101,991
#impressions	230,117	15,777,377	-

Overall Performance

	Method	MIND-small				MIND-large			
		AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
Neural NewsRec Methods	NRMS	65.63	30.96	34.13	40.52	68.24	33.49	36.56	42.24
	GERL	65.27	30.10	32.93	39.48	68.10	33.41	36.34	42.03
	HieRec	67.95	32.87	36.36	42.53	69.03	33.89	37.08	43.01
	PLM-NR	68.60	32.97	36.55	42.78	69.50	34.75	37.99	43.72
	UNBERT	67.92	31.72	34.75	41.02	70.68	<u>35.68</u>	<u>39.13</u>	44.78
	DIGAT	68.77	33.46	37.14	<u>43.39</u>	70.08	35.20	38.46	44.15
	PUNR	68.89	33.33	36.94	43.10	<u>71.03</u>	35.17	39.04	<u>45.41</u>
	TDNR-C ²	68.89	33.57	37.23	<u>43.39</u>	70.38	34.62	38.12	44.30
Knowledge-aware NewsRec Methods (More using entity, KG)	DKN	62.90	28.37	30.99	37.41	64.07	30.42	32.92	38.66
	KRED	65.33	30.60	33.42	39.98	68.52	33.78	36.76	42.45
	UaG	65.10	29.89	33.31	39.46	69.23	34.14	37.21	43.04
	GREP	68.12	<u>33.75</u>	<u>37.25</u>	43.37	69.44	34.40	37.54	43.22
	FUM	67.11	31.31	35.08	41.42	70.01	34.51	37.68	43.38
	PerCoNet [†]	<u>68.93</u>	33.40	36.93	43.28	-	-	-	-
	GLORY	68.15	32.97	36.47	42.78	69.45	34.03	37.92	44.19
	IP2	69.69*	34.51*	38.30*	44.42*	72.06*	35.96*	40.09*	46.35*

- ❑ Compare with entity/KG-based methods and methods using rich news item informations
 - UNBERT: models word-level intra-/inter-news interests
 - GREP: enriches title entities with neighboring KG entities
- ❑ IP2 effectively captures the interaction between entity-based and title-based interest streams

Overall Performance

	Method	MIND-small				MIND-large			
		AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
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	HieRec	67.95	32.87	36.36	42.53	69.03	33.89	37.08	43.01
	PLM-NR	68.60	32.97	36.55	42.78	69.50	34.75	37.99	43.72
	UNBERT	67.92	31.72	34.75	41.02	70.68	35.68	39.13	44.78
	DIGAT	68.77	33.46	37.14	43.39	70.08	35.20	38.46	44.15
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	KRED	65.33	30.60	33.42	39.98	68.52	33.78	36.76	42.45
	UaG	65.10	29.89	33.31	39.46	69.23	34.14	37.21	43.04
	GREP	68.12	33.75	37.25	43.37	69.44	34.40	37.54	43.22
	FUM	67.11	31.31	35.08	41.42	70.01	34.51	37.68	43.38
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	IP2	69.69*	34.51*	38.30*	44.42*	72.06*	35.96*	40.09*	46.35*

Neural NewsRec Methods



Method	Adressa-1week			
	AUC	MRR	nDCG@5	nDCG@10
NRMS	75.31	42.24	44.66	48.46
HieRec	78.67	49.22	48.72	56.67
PLM-NR	78.20	47.26	48.41	54.60
PUNR	78.32	47.71	49.32	54.80
IP2	83.16*	50.83*	54.05*	59.32*

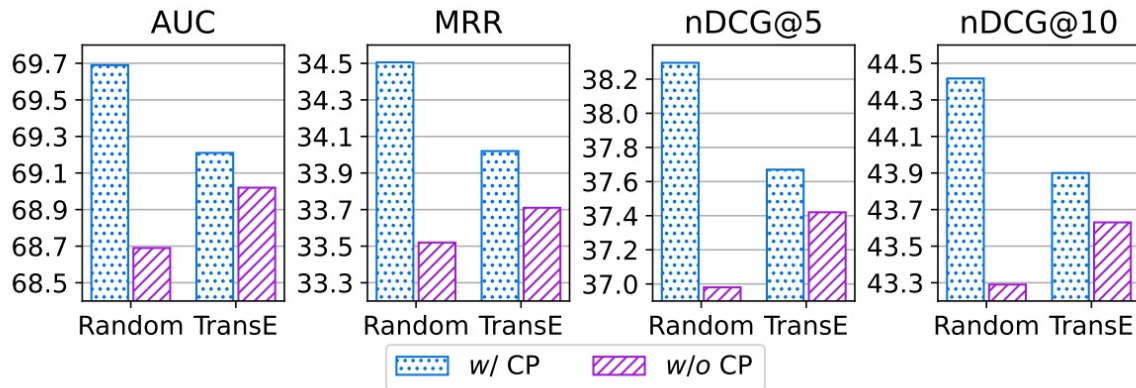
❑ All methods that explicitly utilize entity embeddings learned from KG do not work on Adressa-1week

- ▶ Lack of a comprehensive knowledge graph in Norwegian
- ▶ While IP2 acquire entity representations without KG

❑ Fine-grained guidance is essential in news recommendations

- ▶ Both PUNR and PLM-NR are inferior to HieRec
 - Average title length, MIND: 10.79 vs Adressa: 6.57
 - Limited semantics restricts interest modeling capability of text-only methods

Analysis on Contrastive Pre-training



- ❑ Comparison between initializing entity memory with Random values / TransE-Wikidata embeddings
 - Random + CP -> IP2's default setup
- ❑ Both initialization method improved performance with contrastive pre-training
- ❑ Using TransE embeddings may introduce unexpected noise that hinders performance
 - Random + CP > TransE + CP
 - Due to the aforementioned inherent issues with KG

Case Study

- 7 user history news, 9 candidate news, clicked 2 news
 - Compared with UNBERT, GREP
- During contrastive pre-training,
 - “Prince Harry” and “Meghan Markle” appear together multiple times
 - recognizes them as a couple within the royal family from the context
- IP2 further probes “Prince Harry” in N20953 as the intra-news entity focus
 - Both IP2 and GREP are capable of capturing the entity interest between N20953 and N21325
 - Literal meaning between N20953 and N21325 differs so much
 - GREP finally ranks N21325 to #4

User U89104's reading history

ID	Title
N11723	'Wheel Of Fortune' Guest Delivers Hilarious, Off The Rails Introduction
N26330	Felicity Huffman Is Scheduled to Be Released from Prison on October 27 After Serving 13 Days
N20723	Maleficent: Mistress of Evil delivers an empty, CG-dazzled fairytale
N20953	Prince Harry acknowledges tensions with William in ITV interview
N14065	Heidi Klum's 2019 Halloween Costume Transformation Is Mind-Blowing But, Like, What Is It?
N27591	Pamela Anderson gets backlash after wearing a Native American headdress for Halloween
N425	Here Are the Biggest Deals We're Anticipating for Black Friday

Recommendation (rank out of 9 candidate news items)

ID	Candidate News	IP2	UNBERT	GREP
N21325	Meghan Markle and Hillary Clinton Secretly Spent the Afternoon Together at ...	1	6	4
N20150	30 Best Black Friday Deals from Costco	2	1	3

Other news articles in the dataset

ID	Title
N12157	Prince Harry & Meghan Markle's Trusted Private Secretary Has Resigned
N4117	Prince Harry and Meghan Markle are Taking a Royal Respite

Case Study (Cont.)

- ❑ User U89104 is interested in “Black Friday” based on reading history
 - “Black Friday” is not labeled as an entity in N425
- ❑ For UNBERT, since there is a direct word match, N20150 is ranked as #1
 - While GREP rank N20150 to #3
 - Entity missing affects the user encoder in GREP
- ❑ IP2’s cross attention link bridges these two “black fridays” together
 - Catch interest from title semantic and link it
 - IP2 ranks N20150 to position #2

User U89104’s reading history

ID	Title
N11723	'Wheel Of Fortune' Guest Delivers Hilarious, Off The Rails Introduction
N26330	Felicity Huffman Is Scheduled to Be Released from Prison on October 27 After Serving 13 Days
N20723	Maleficent: Mistress of Evil delivers an empty, CG-dazzled fairytale
N20953	Prince Harry acknowledges tensions with William in ITV interview
N14065	Heidi Klum's 2019 Halloween Costume Transformation Is Mind-Blowing But, Like, What Is It?
N27591	Pamela Anderson gets backlash after wearing a Native American headdress for Halloween
N425	Here Are the Biggest Deals We're Anticipating for Black Friday

Recommendation (rank out of 9 candidate news items)

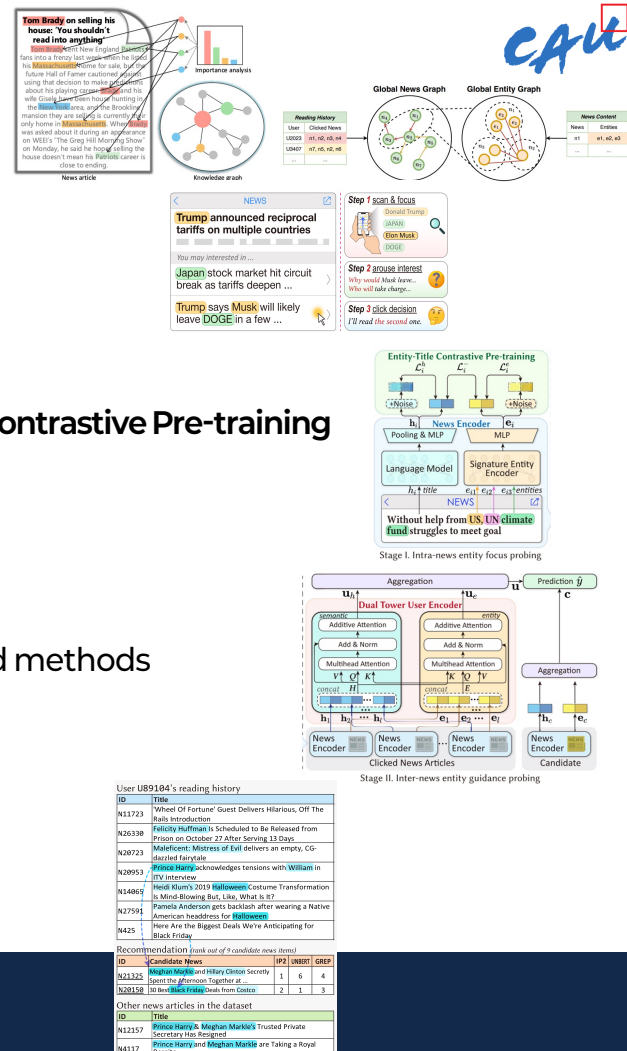
ID	Candidate News	IP2	UNBERT	GREP
N21325	Meghan Markle and Hillary Clinton Secretly Spent the Afternoon Together at ...	1	6	4
N20150	30 Best Black Friday Deals from Costco	2	1	3

Other news articles in the dataset

ID	Title
N12157	Prince Harry & Meghan Markle's Trusted Private Secretary Has Resigned
N4117	Prince Harry and Meghan Markle are Taking a Royal Respite

Conclusion

- Existing methods that treat entities as mere semantic features, overlooking entities' unique roles on guiding interest stream
- IP2 captures intra-, inter- news entity guided interest stream
 - Intra-news entity focus probing with Signature Entity Encoder, Entity-Title Contrastive Pre-training
 - Intra-news entity focus probing with Dual Tower User Encoder
 - Learnable aggregation to capture different user behavior patterns
- Experiments compare IP2 with Neural methods and Entity, Knowledge-based methods
 - Show robust performance both Datasets, MIND and Adressa
- Analysis on Contrastive Pre-training without KG-based entity initialization
- Case study analyzes detailed recommendation flows and compares them with other models





Modeling Stage-wise Evolution of User Interests for News Recommendation

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Information Engineering, Hefei
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Published in WWW' 2026

Timestamp in News Recommendation Data

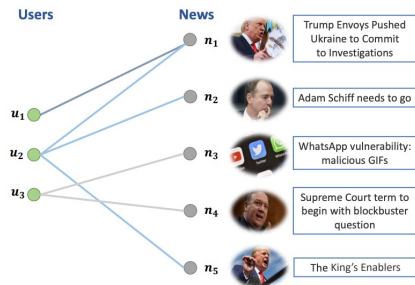
❖ Adressa Dataset

Attribute	Description	Example
eventID	Id of reading event	1082287123 (integer)
Time	The time of the event	1487572383 (Unix time)
sessionStart	Indicates if the event is the first event in the session	False
sessionStop	Indicates if the event is the last event in the session	False
activeTime	The active time spent on a page	23 (seconds)
canonicalURL	URL of the visited page	"http://adressa.no"
referrerURL	The URL of the referrer page	"www.facebook.com"
documentId	Internal ID of page	"9757814edc2d346dfcf6f54e349f404c4e9775cf"
Title	Title of the article	"Test av 19 grovbrød"
Category	News category	"sport"

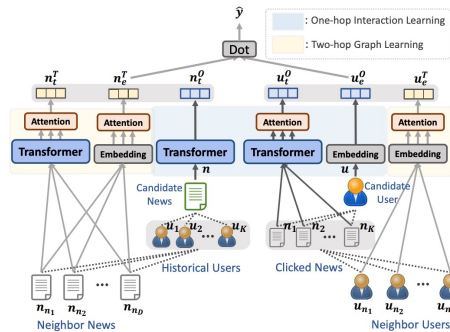
- The dataset provides timestamp information indicating **when each user–news interaction occurred**

Previous Approaches on User–News Graphs

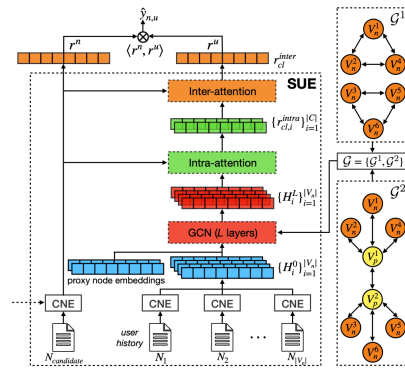
❖ GCN-based Encoders for Static User–News or History Graphs



User-News Bipartite Graph



GERL (WWW' 20)



CNE-SUE (EMNLP' 21)

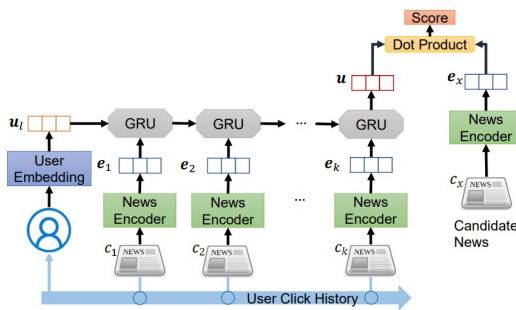
❑ Capture structural and collaborative signals

- ▶ Utilize high-order connectivity to improve representation consistency

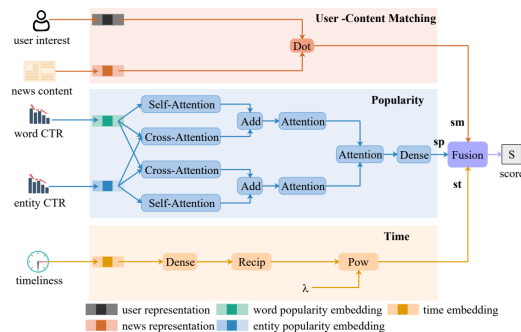
➔ Assume user preferences are static and fail to capture how they evolve over time

Previous Approach on Time-aware Designs

❖ The Challenge of Balancing Short-term Sensitivity with Long-term Stability



LSTUR (ACL' 19)

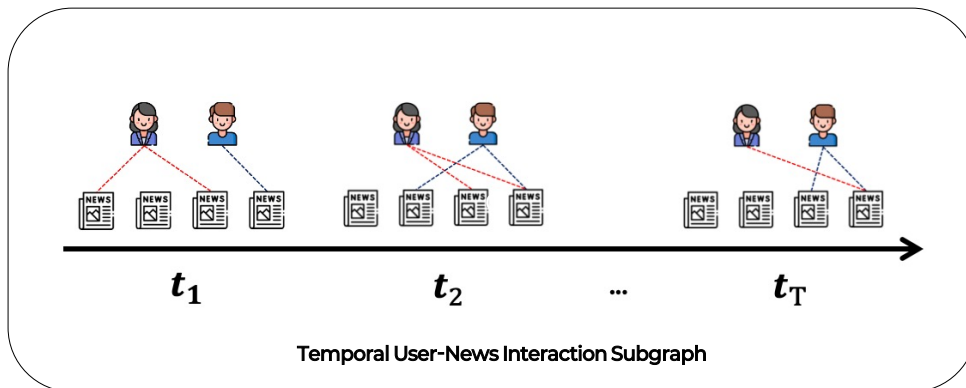
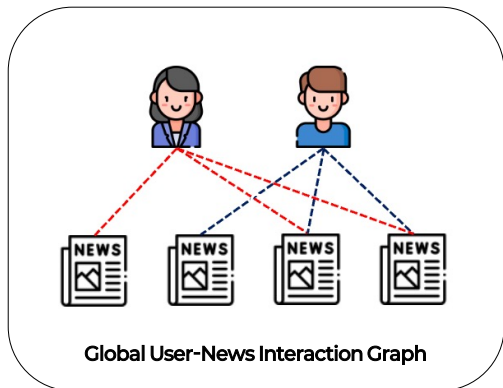


TCCM (CIKM' 23)

- ❑ Reflect temporal information using click order and news publication time
- ➔ May be dominated by outdated interactions
- ➔ May become biased toward short-term interests

Global Collaborative & Time-dependent Signal

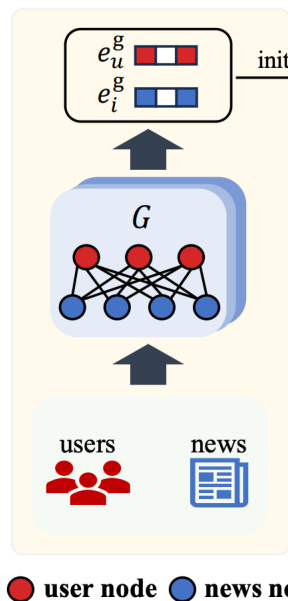
❖ Model with Global Graph and Temporal Subgraph



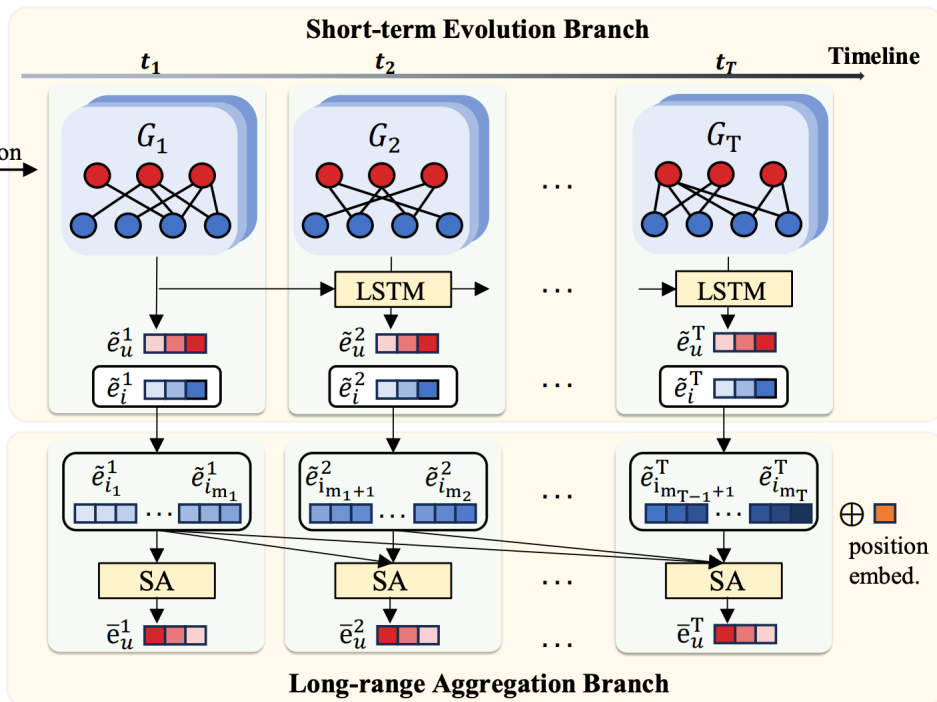
- ❑ Represents user-news interactions through the **time stage subgraph structure**
 - Clicks within the same period often reflect coherent local interests
 - While transitions across periods capture **how preferences evolve over time**
- ❑ Capture both short-term interests and long-range preferences on these graphs

GPM-LPM (Global Preference Modeling & Local Preference Modeling)

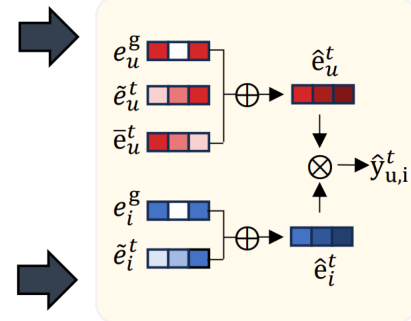
Global Preference Modeling



Local Preference Modeling

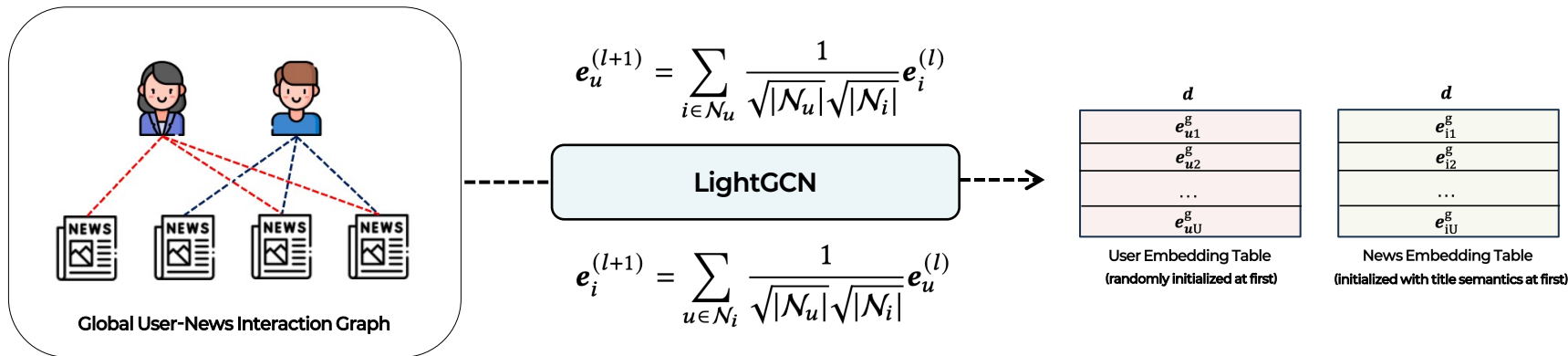


- position embed.
- SA self-attention
- $\oplus$ element-wise add.
- $\otimes$ element-wise mult.



Global Preference Modeling

❖ Learn Global User and News Embeddings with LightGCN

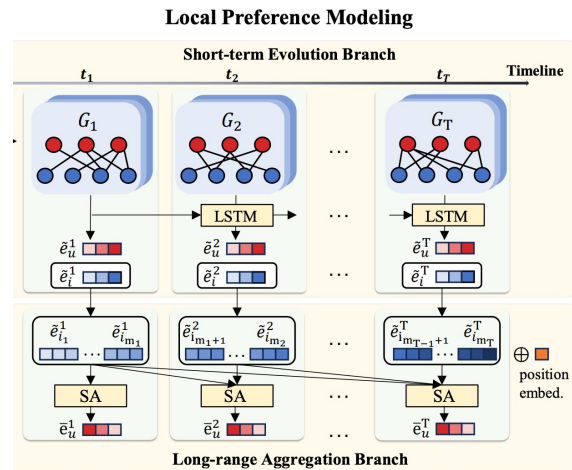


- ❑ Global preference modeling captures long-term preferences
 - Reflect the **relatively stable interests of the users** accumulated over a long history of interactions
- ❑ Global embeddings e_u^g and e_i^g are **used to initialize embedding learning in temporal subgraphs**
 - High-quality initialization that facilitates more effective embedding learning in the temporal modules

Local Preference Modeling

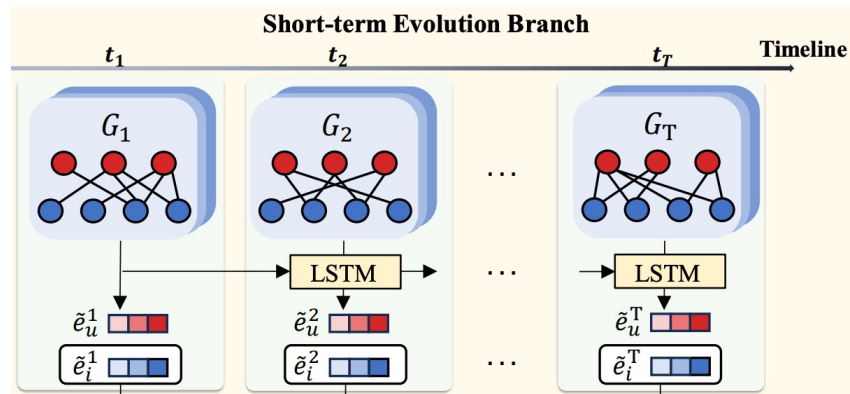
❖ Two Complementary Branches for Capturing User Interest Evolution

- ❑ Short-term Evolution Branch
 - Using LSTM to capture progressive and recency-sensitive preference shifts
- ❑ Long-range Aggregation Branch
 - Employing self-attention to model cross-stage dependencies and highlight historically influential signals
- ❑ Capture both short-term dynamics and long-range dependencies



Short-term Evolution Branch

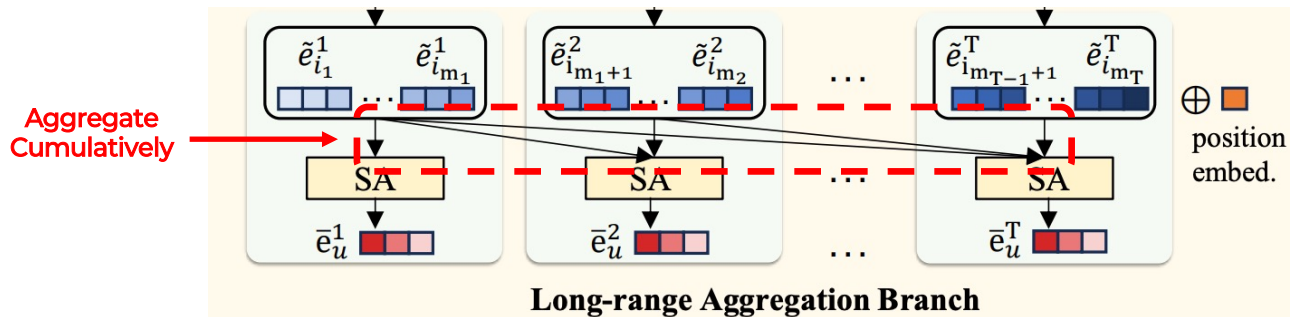
- ❖ Learn User and News Representations from Each Temporal Subgraph



- Stage-specific representations preserve the unique semantics of each time window
 - $\mathbf{e}_u^t, \mathbf{e}_i^t = \text{GCN}^{(t)}(\mathcal{G}_t; \mathbf{e}_u^g, \mathbf{e}_i^g), t = \{1, \dots, T\}.$
- Model sequential dependencies to capture preferences evolution across adjacent stages
 - $\tilde{\mathbf{e}}_u^t = \text{LSTM}(\mathbf{e}_u^t, \mathbf{e}_u^{(t-1)}), \quad \tilde{\mathbf{e}}_i^t = \text{LSTM}(\mathbf{e}_i^t, \mathbf{e}_i^{(t-1)})$

Long-range Aggregation Branch

❖ Consider Cumulative Click Histories Across Temporal Stages



- ❑ Capture stage-wise evolution by adaptively weighting historical and recent signals
 - Apply self-attention to accumulated clicked news sequences
 - Add learnable positional encodings as chronological anchors
- ❑ Sum the L-self-attention layers outputs over clicked-news positions
 - Obtain a time-indexed cumulative user representation up to stage t

Objective Function

❖ Prediction

- Final representation of user, news

$$\hat{\mathbf{e}}_u^t = \underset{\text{short}}{\tilde{\mathbf{e}}_u^t} + \underset{\text{long}}{\bar{\mathbf{e}}_u^t} + \underset{\text{global}}{\mathbf{e}_u^g}, \quad \hat{\mathbf{e}}_i^t = \underset{\text{short}}{\tilde{\mathbf{e}}_i^t} + \underset{\text{global}}{\mathbf{e}_i^g}.$$

- Scoring function

$$\underset{\text{sigmoid}}{\hat{y}_{u,i}^t} = \sigma(\hat{\mathbf{e}}_u^t \cdot \hat{\mathbf{e}}_i^t), \text{ , larger values indicating a higher probability of user-item interaction}$$

❖ Prediction loss (BCE Loss)

$$\square L_t = - \sum_{(u,i) \in \mathcal{D}_t} \left[y_{u,i}^t \log \hat{y}_{u,i}^t + (1 - y_{u,i}^t) \log(1 - \hat{y}_{u,i}^t) \right],$$

Regularization

❖ Consistency Regularization

- Align with the global preference patterns learned from the complete interaction graph

- $$L_{\text{cl}}^t = -\log \frac{\exp(\tilde{\mathbf{e}}_u^t \cdot \mathbf{e}_u^g / \tau)}{\sum_j \exp(\tilde{\mathbf{e}}_u^t \cdot \mathbf{e}_j^g / \tau)}, \quad L_{\text{cl}} = \sum_{t=1}^T L_{\text{cl}}^t.$$

❖ Smoothness Regularization

- Enforce smooth transitions between adjacent temporal embeddings

- $$L_{\text{sl}}^t = \|\tilde{\mathbf{e}}_u^t - \tilde{\mathbf{e}}_u^{(t-1)}\|_2^2, \quad L_{\text{sl}} = \sum_{t=1}^T L_{\text{sl}}^t.$$

❖ Final Optimization Objective

- $$L = \sum_{t=1}^T \lambda_t L_t + \lambda_{\text{cl}} L_{\text{cl}} + \lambda_{\text{sl}} L_{\text{sl}} + \beta \|\Theta\|_2^2.$$

Experiment

Table 1: Statistics of news article datasets.

Dataset	Users	News	Clicks	Time Span
Adressa-large	3,614,911	81,018	35,244,078	3 months
MIND-large	1,000,000	161,013	24,155,470	6 weeks



❖ Experimental Setup

- ❑ Adressa-large provides complete timestamps in the official release
 - Set the window size to two weeks
- ❑ MIND dataset, weeks 1–4 are chronologically ordered but do not provide exact timestamps
 - Set the window size to one week (divide in quarter)
 - Using the entire week-5 as the test set
 - While earlier studies typically evaluate on a much shorter temporal horizon
 - e.g., only the final day of week-5

❖ Metrics

- ❑ AUC, MRR, NDCG@5, NDCG@10

Overall Performance

Dataset	Metric	NPA	NRMS	NAML	LightGCN	CNE-SUE	TCCM	CROWN	Ours
Adressa	AUC	0.5312	0.5091	0.5046	0.4996	0.5207	0.5419	<u>0.6553</u>	0.7993
	MRR	0.2521	0.2512	0.2602	0.4056	0.2471	0.3424	<u>0.3915</u>	0.5223
	NDCG@5	0.2263	0.2206	0.2341	0.2812	0.2209	0.2380	<u>0.4156</u>	0.4495
	NDCG@10	0.3103	0.3018	0.3002	0.3379	0.3046	0.3219	<u>0.5283</u>	0.5327
MIND	AUC	0.5008	0.5429	0.5479	0.5029	0.5266	0.5468	<u>0.5501</u>	0.5804
	MRR	0.2947	0.3227	0.3173	0.2491	0.2366	0.3570	<u>0.3431</u>	0.4368
	NDCG@5	0.3843	0.3566	0.3489	0.2289	0.2458	0.3178	<u>0.4156</u>	0.4429
	NDCG@10	0.4176	0.4218	0.4134	0.2908	0.3077	0.4068	<u>0.4691</u>	0.5171

- Show robustness result with different temporal spans and interaction characteristics
 - Models that fail to capture temporal evolution or rely on static graphs show weaker accuracy
 - CROWN, the second best model, its dependence on explicit content features restricts adaptability

Ablation Study

❑ w/o LPM

- Only use Global Preference Modeling

❑ w/o STE

- Without LSTM in short-term evolution branch'
- Adapt to rapidly shifting news interests
approve metrics sensitive to recency (ranking metrics)

❑ w/o LRA

- Only use STE in Local Preference Modeling (without self-attention)
- Incorporate non-local historical context significantly is also important

❑ w/o GPM

- Without high-order collaborative patterns, informative initialization

Variant	AUC	MRR	N@5	N@10
w/o LPM	0.4880	0.1885	0.1049	0.1416
w/o STE	0.5109	0.2112	0.2570	0.2713
w/o LRA	0.4851	0.2573	0.3120	0.3322
w/o GPM	0.4996	0.4056	0.2812	0.3379
Ours	0.7993	0.5223	0.4495	0.5327

Model Capability

□ Evaluate the model can capture the temporal evolution of user interests

- Group: divide users based on interaction frequency
- New: published within the most recent one-month
- Historical: published during the earliest one-month

□ Show a better balance between novelty and relevance

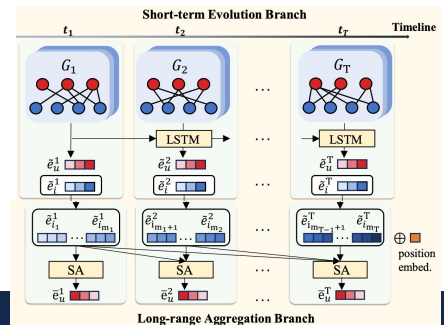
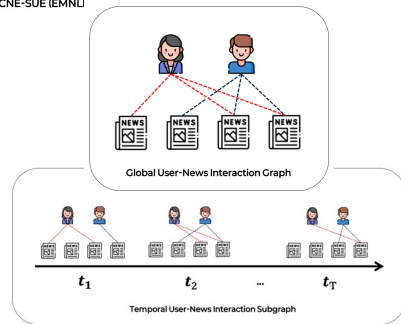
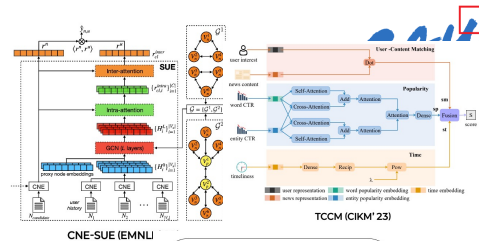
- In the Medium, Low users, LPM increases the share of new items, while improving the ranking of historical articles

Group	Model	Proportion in top-10		New	Old
		New%	Historical%	NRank	ORank
High	w/o LPM	0.035	0.036	1.59	1.58
	our model	0.014	0.018	0.60	0.82
Med	w/o LPM	0.164	0.161	4.31	4.32
	our model	0.187↑	0.145↓	4.53↓	3.85↑
Low	w/o LPM	0.159	0.156	4.10	4.12
	our model	0.173↑	0.148↓	4.09↑	3.74↑
Rand	w/o LPM	0.114	0.113	3.38	3.40
	our model	0.122	0.102	3.28	3.04

- ****I don't think this Exp shows *"Performance of capturing temporal evolution of user interests in detail"* but just shows that model isn't biased at each side of timeliness, balancing performance.****

Conclusion

- ❑ Using "static" user-news graph assume user preference to be static and fail to capture how they evolve over time
- ❑ Previous time-aware design might be dominated by outdated interactions or become biased toward short-term interests
- ❑ GPM-LPM use both global User-News Interaction Graph and Temporal Subgraph
 - Captures long-term stable, collaborative preference signal on Global Graph
 - Using GCN & LSTM
 - Captures time-sensitive short-term interests and long-range dependencies on Temporal Subgraph
 - Using Cumulative Self-Attention Layer
- ❑ Experiment shows robustness result with different temporal spans and interaction characteristics and contribution of each module of GPM-LPM



Appendix (Ablation Study, IP2)

□ w/o Intra (-news entity guidance)

- Removes entity-title contrastive pre-training
 - No initialize on entity embedding
- Partially remove the entity-title part, w/o CL_{e-t} , w/o CL_{t-t} & $e-e$
 - CL_{e-t} directly affects whether entity memory could be initialized with proper meanings

$$\mathcal{L}_i^- = -\log \frac{\exp(\text{sim}(\mathbf{e}_i, \mathbf{h}_i)/\tau)}{\sum_{j=1}^v \exp(\text{sim}(\mathbf{e}_i, \mathbf{h}_j)/\tau)},$$

$$CL_{e-t}$$

$$\mathcal{L}_i^h = -\log \frac{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau)}{\sum_{j=1}^v \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)/\tau)},$$

$$\mathcal{L}_i^e = -\log \frac{\exp(\text{sim}(\mathbf{e}_i, \mathbf{e}_i^+)/\tau)}{\sum_{j=1}^v \exp(\text{sim}(\mathbf{e}_i, \mathbf{e}_j^+)/\tau)}.$$

$$CL_{t-t \& e-e}$$

□ w/o Inter (-news entity guidance)

- Removes cross attention and uses self-attention in each user encoder tower
- Inter-news stream-wise interaction contributes mostly to the users' reading decision

□ w/o Agg

- without aggregation layer concatenates u_h and u_e , h_c and e_c without using learnable weight
 - just concatenate two embeddings
- entity- and semantic-guided reading preferences may hold varying importance for different users

Variant	AUC	MRR	N@5	N@10
IP2 (original)	69.69*	34.51*	38.30*	44.42*
w/o Intra	68.69	33.52	36.98	43.29
w/o CL_{e-t}	68.63	33.70	37.16	43.37
w/o $CL_{t-t \& e-e}$	69.03	34.01	37.58	43.80
w/o Inter	68.66	33.28	36.65	43.04
w/o Agg	68.44	33.38	36.87	43.12

Appendix (Compared Model, IP2)

□ NRMS <-> HieRec, NRMS <-> PLM-NR

- ▶ topic, subcategory를 활용하는 fine-grained interest modeling
- ▶ boosting performance by providing semantic info. using PLM

□ Entity utilized methods (PerCoNet, GLORY)

- ▶ PerCoNet: Using entity-based personality analysis on user encoder in PLM-NR
- ▶ GLORY: Using global/local entity graph for enhancing user behavior modeling

□ UNBERT

- ▶ Modeling intra- inter-news interest on word-level
 - unnecessary words may bring noise

Method	MIND-small				MIND-large			
	AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
NRMS	65.63	30.96	34.13	40.52	68.24	33.49	36.56	42.24
GERL	65.27	30.10	32.93	39.48	68.10	33.41	36.34	42.03
HieRec	67.95	32.87	36.36	42.53	69.03	33.89	37.08	43.01
PLM-NR	68.60	32.97	36.55	42.78	69.50	34.75	37.99	43.72
UNBERT	67.92	31.72	34.75	41.02	70.68	35.68	39.13	44.78
DIGAT	68.77	33.46	37.14	43.39	70.08	35.20	38.46	44.15
PUNR	68.89	33.33	36.94	43.10	71.03	35.17	39.04	45.41
TDNR-C ²	68.89	33.57	37.23	43.39	70.38	34.62	38.12	44.30
DKN	62.90	28.37	30.99	37.41	64.07	30.42	32.92	38.66
KRED	65.33	30.60	33.42	39.98	68.52	33.78	36.76	42.45
UaG	65.10	29.89	33.31	39.46	69.23	34.14	37.21	43.04
GREP	68.12	33.75	37.25	43.37	69.44	34.40	37.54	43.22
FUM	67.11	31.31	35.08	41.42	70.01	34.51	37.68	43.38
PerCoNet [†]	68.93	33.40	36.93	43.28	-	-	-	-
GLORY	68.15	32.97	36.47	42.78	69.45	34.03	37.92	44.19
IP2	69.69*	34.51*	38.30*	44.42*	72.06*	35.96*	40.09*	46.35*

Appendix (Compared Model, Cont.)

□ FUM

- Tries to model entity-guided interest
 - But overlooks the intra-news level

□ GRep

- utilizes an entity-dedicated user encoder
 - GNN backbone is prone to encountering the cold-start problem on the test set

□ PUNR

- remarkable results on MIND-large
 - BERT-like pre-training task demands a significant amount of interaction logs
 - making it suboptimal on MIND-small

Method	MIND-small				MIND-large			
	AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
NRMS	65.63	30.96	34.13	40.52	68.24	33.49	36.56	42.24
GERL	65.27	30.10	32.93	39.48	68.10	33.41	36.34	42.03
HicRec	67.95	32.87	36.36	42.53	69.03	33.89	37.08	43.01
PLM-NR	68.60	32.97	36.55	42.78	69.50	34.75	37.99	43.72
UNBERT	67.92	31.72	34.75	41.02	70.68	35.68	39.13	44.78
DIGAT	68.77	33.46	37.14	43.39	70.08	35.20	38.46	44.15
PUNR	68.89	33.33	36.94	43.10	71.03	35.17	39.04	45.41
TDNR-C ²	68.89	33.57	37.23	43.39	70.38	34.62	38.12	44.30
DKN	62.90	28.37	30.99	37.41	64.07	30.42	32.92	38.66
KRED	65.33	30.60	33.42	39.98	68.52	33.78	36.76	42.45
UaG	65.10	29.89	33.31	39.46	69.23	34.14	37.21	43.04
GREP	68.12	33.75	37.25	43.37	69.44	34.40	37.54	43.22
FUM	67.11	31.31	35.08	41.42	70.01	34.51	37.68	43.38
PerCoNet [†]	68.93	33.40	36.93	43.28	-	-	-	-
GLORY	68.15	32.97	36.47	42.78	69.45	34.03	37.92	44.19
IP2	69.69*	34.51*	38.30*	44.42*	72.06*	35.96*	40.09*	46.35*

Appendix (Hyperparameter, GPM-LPM)

- Only test on Adressa dataset
 - provides complete timestamps in the official release
- Report the results using AUC
 - other metrics follow similar trends

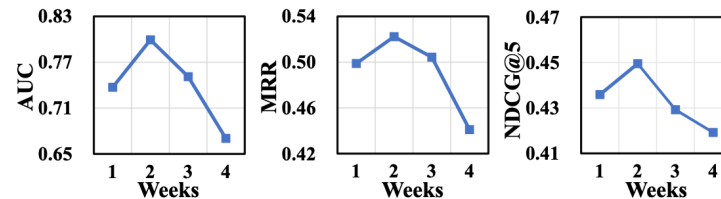
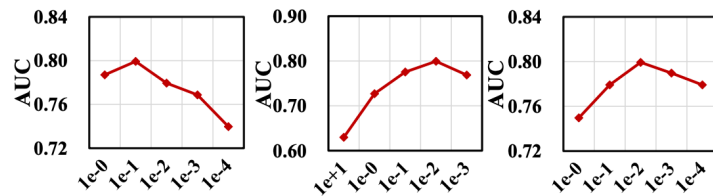


Figure 2: Performance under different window size.



(a) Prediction λ_t (b) Consistency λ_{cl} (c) Smoothness λ_{sl}

Figure 3: Effect of loss weight coefficients on AUC.

Appendix (NDCG)

Rank	News	Click 여부
1	A	0
2	B	0
3	C	1 (정답)
4	D	0
5	E	1 (정답)

$$DCG = \frac{1}{\log_2(\text{rank} + 1)}$$

- DCG@10 =

$$\frac{1}{\log_2(3 + 1)} + \frac{1}{\log_2(5 + 1)} = 0.5 + 0.386 = 0.886$$

- IDCG@10 (정답이 1위, 2위였다고 가정)

$$1 + \frac{1}{\log_2(3)} = 1 + 0.6309 = 1.6309$$

- NDCG@10 =

$$0.886 / 1.6309 = 0.543$$